

```

#include <stdio.h>

int main()
{
    int pages[50], frames[10];
    int n, f, i, j, k, fault = 0;
    int found, pos, farthest, next;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter reference string:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i = 0; i < f; i++)
        frames[i] = -1;

    printf("\nPage\tFrames\n");

    for(i = 0; i < n; i++)
    {
        found = 0;

        for(j = 0; j < f; j++)
        {
            if(frames[j] == pages[i])
            {
                found = 1;
                break;
            }
        }

        if(!found)
        {
            pos = -1;

            for(j = 0; j < f; j++)
            {
                if(frames[j] == -1)
                {
                    pos = j;
                    break;
                }
            }
        }
    }
}

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    }

    if(pos == -1)
    {
        farthest = -1;

        for(j = 0; j < f; j++)
        {
            next = n + 1;

            for(k = i + 1; k < n; k++)
            {
                if(frames[j] == pages[k])
                {
                    next = k;
                    break;
                }
            }

            if(next > farthest)
            {
                farthest = next;
                pos = j;
            }
        }
    }

    frames[pos] = pages[i];
    fault++;
}

printf("%d\t", pages[i]);

for(j = 0; j < f; j++)
{
    if(frames[j] == -1)
        printf("- ");
    else
        printf("%d ", frames[j]);
}

printf("\n");
}

printf("\nTotal Page Faults = %d\n", fault);

return 0;

```

}

OUTPUT:

Enter number of pages: 12

Enter reference string:

1 2 3 4 1 2 5 1 2 3 4 5

Enter number of frames: 3

Page	Frames
------	--------

1	1 - -
---	-------

2	1 2 -
---	-------

3	1 2 3
---	-------

4	1 2 4
---	-------

1	1 2 4
---	-------

2	1 2 4
---	-------

5	1 2 5
---	-------

1	1 2 5
---	-------

2	1 2 5
---	-------

3	3 2 5
---	-------

4	4 2 5
---	-------

5	4 2 5
---	-------

Total Page Faults = 7